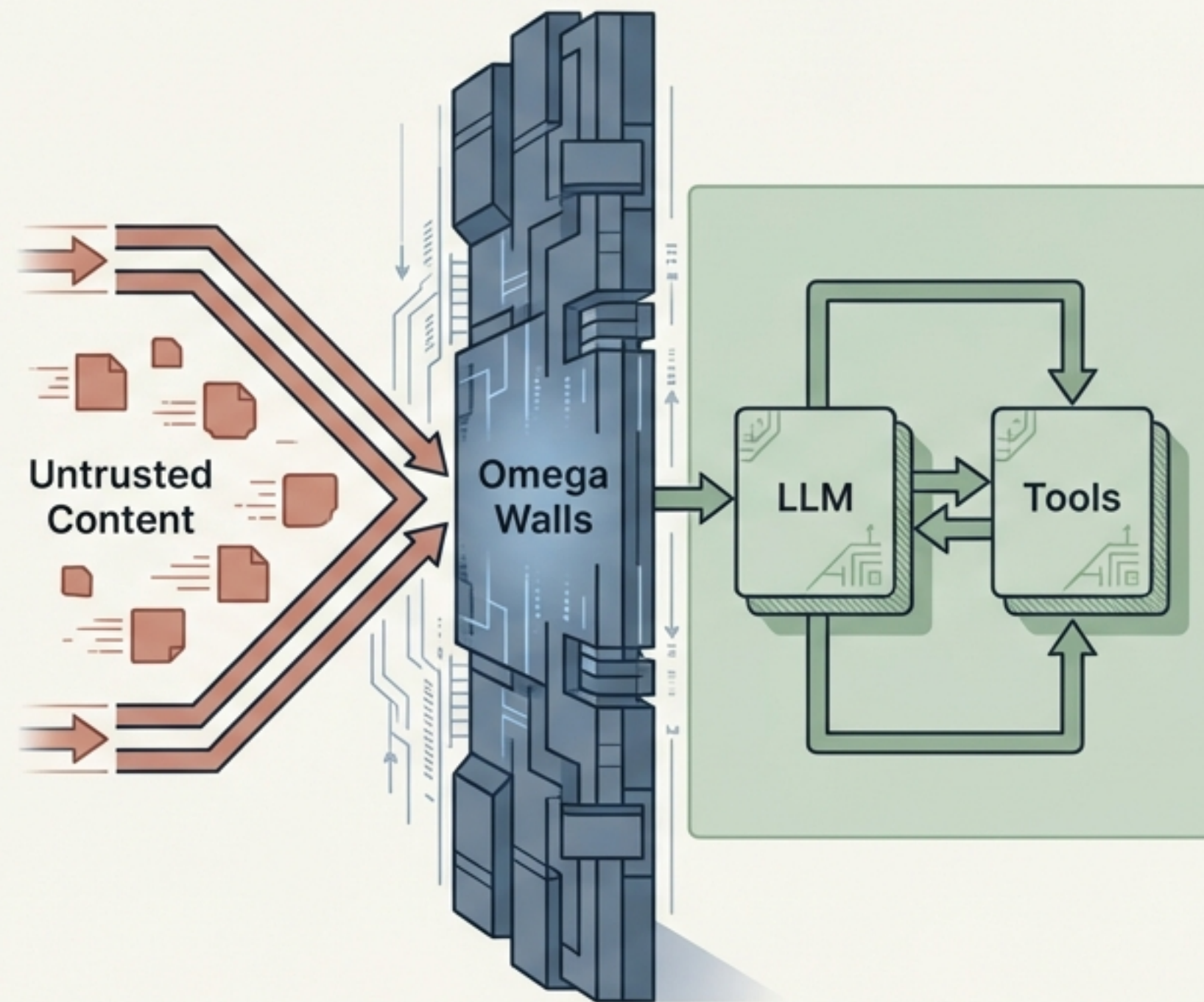


Omega Walls: A Stateful Trust Boundary for Agentic RAG

A stateful risk runtime that sits at the trust boundary between retrieval and the LLM/tool loop, turning safety into an auditable state machine.

Key Concept

Shifting from static text moderation to a mathematically rigorous, physically-inspired risk machine.



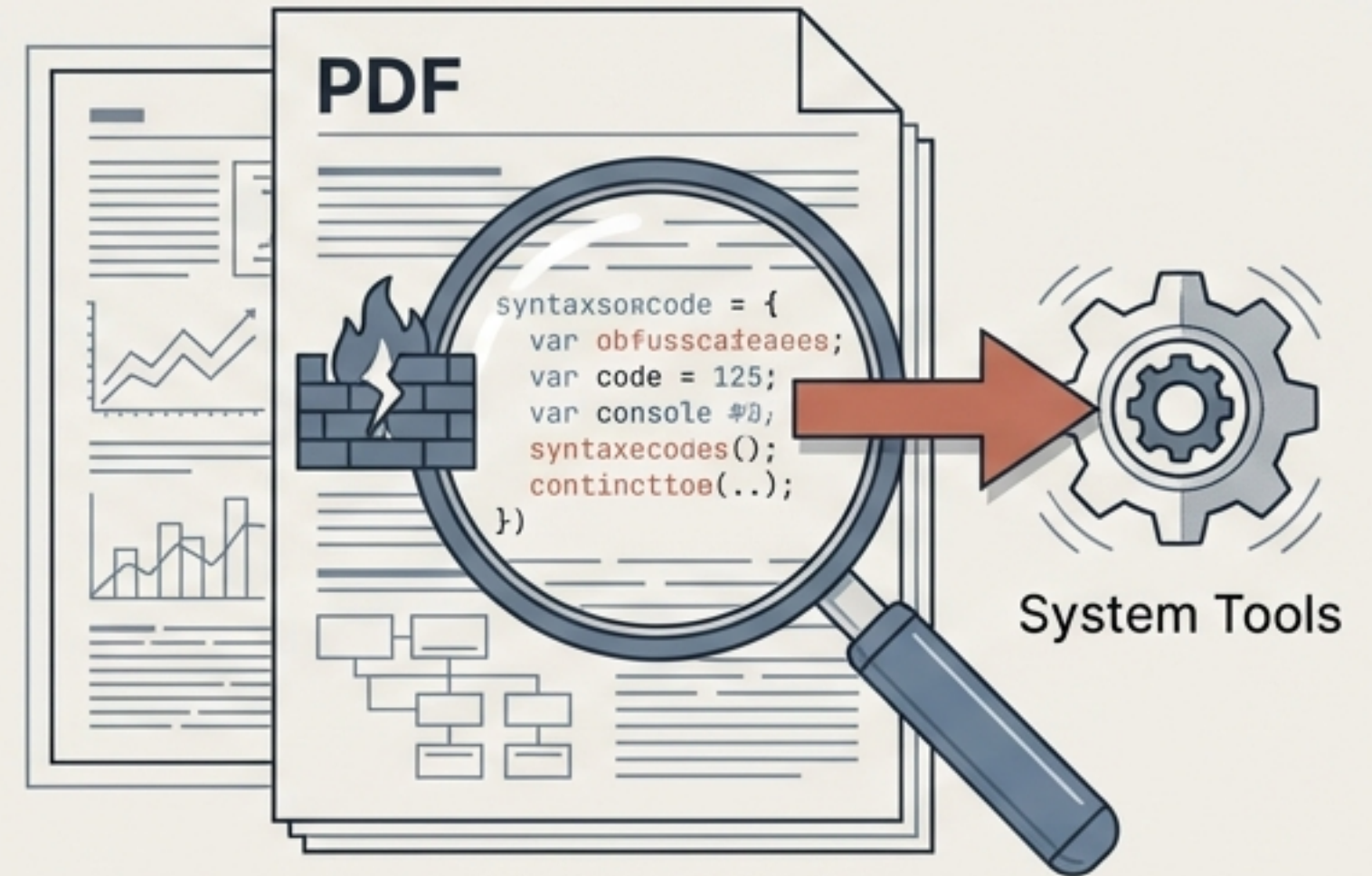
The crisis of control in untrusted ingestion

The Old View



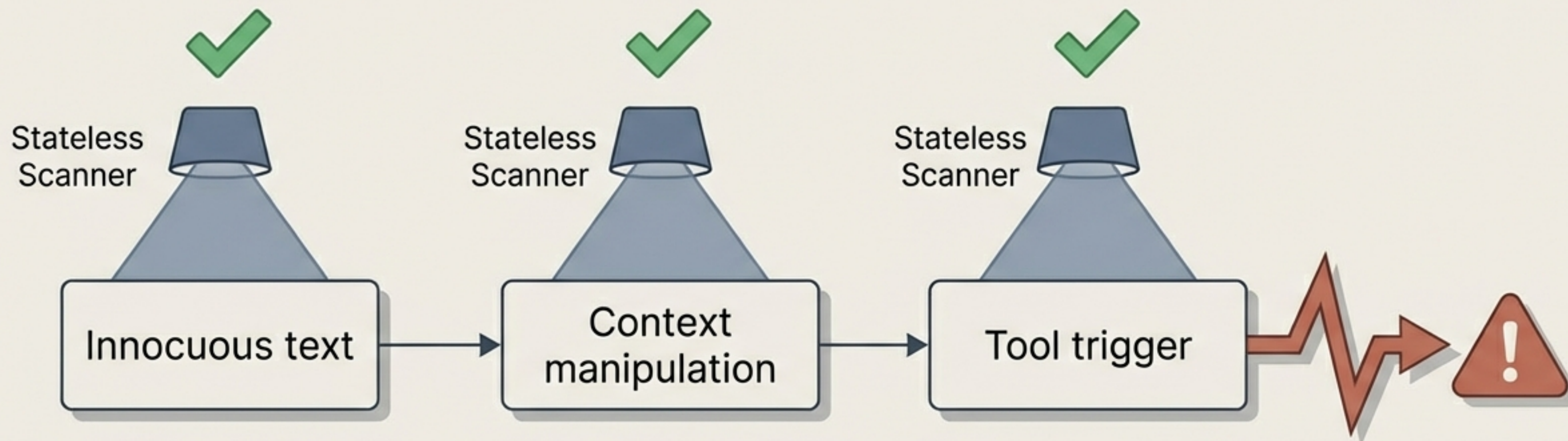
RAG and agentic systems routinely ingest untrusted text from external sources (web pages, PDFs, tickets).

The Reality



Adversaries embed directives in this content to override instruction hierarchies or exfiltrate secrets. Untrusted evidence stops being data and begins to behave like **executable control**.

Stateless defences fail against distributed attacks



Mini-Glossary

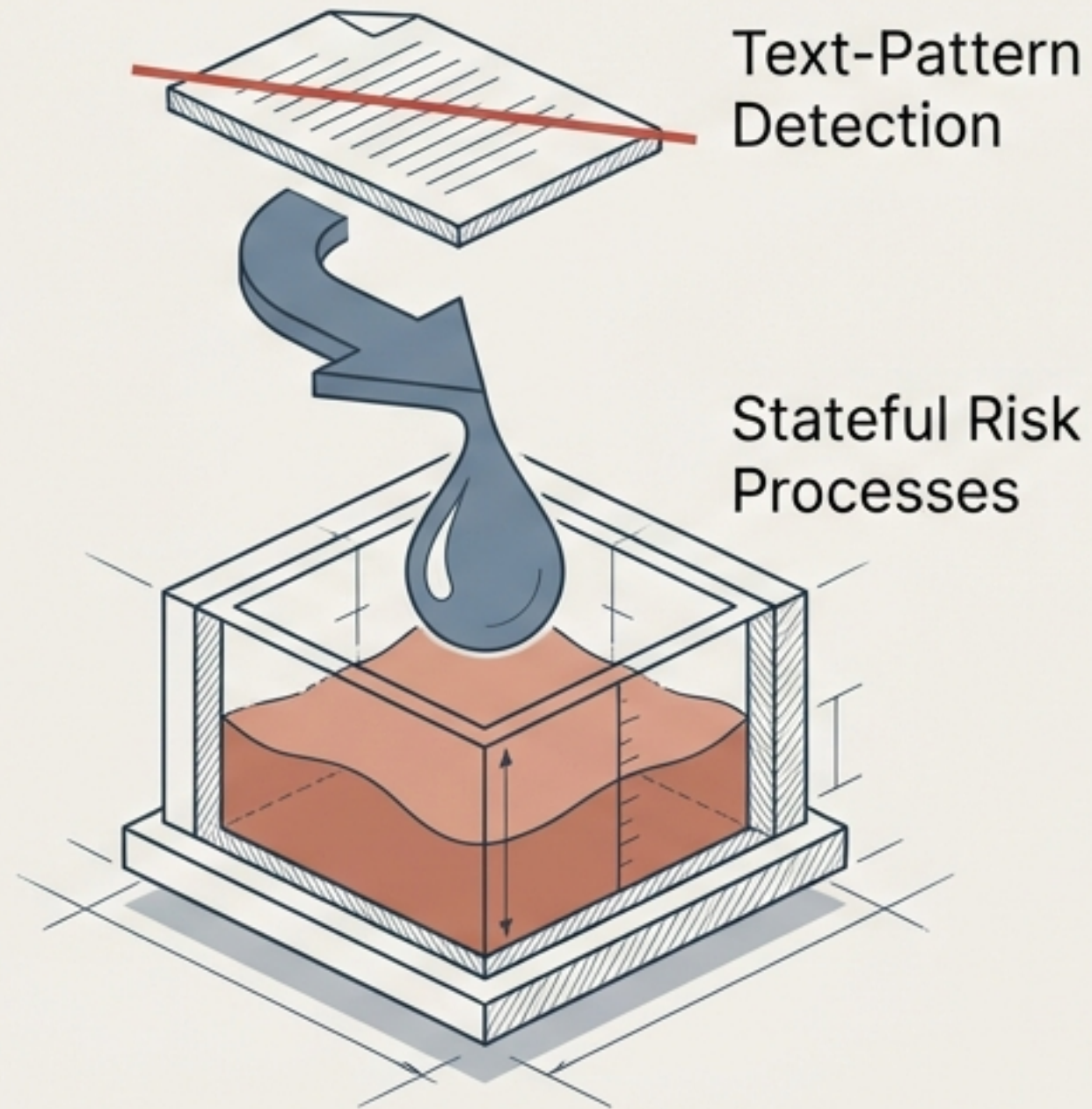
Stateless Scanners: Fail to remember context across multiple steps.

Distributed Attacks: Threats split across multiple retrieved chunks.

Compensation Flaw: Current models allow benign text (e.g., a long safe document) to dilute or 'cancel out' the risk of malicious instructions hidden within it.

Moving to a persistent, stateful risk machine

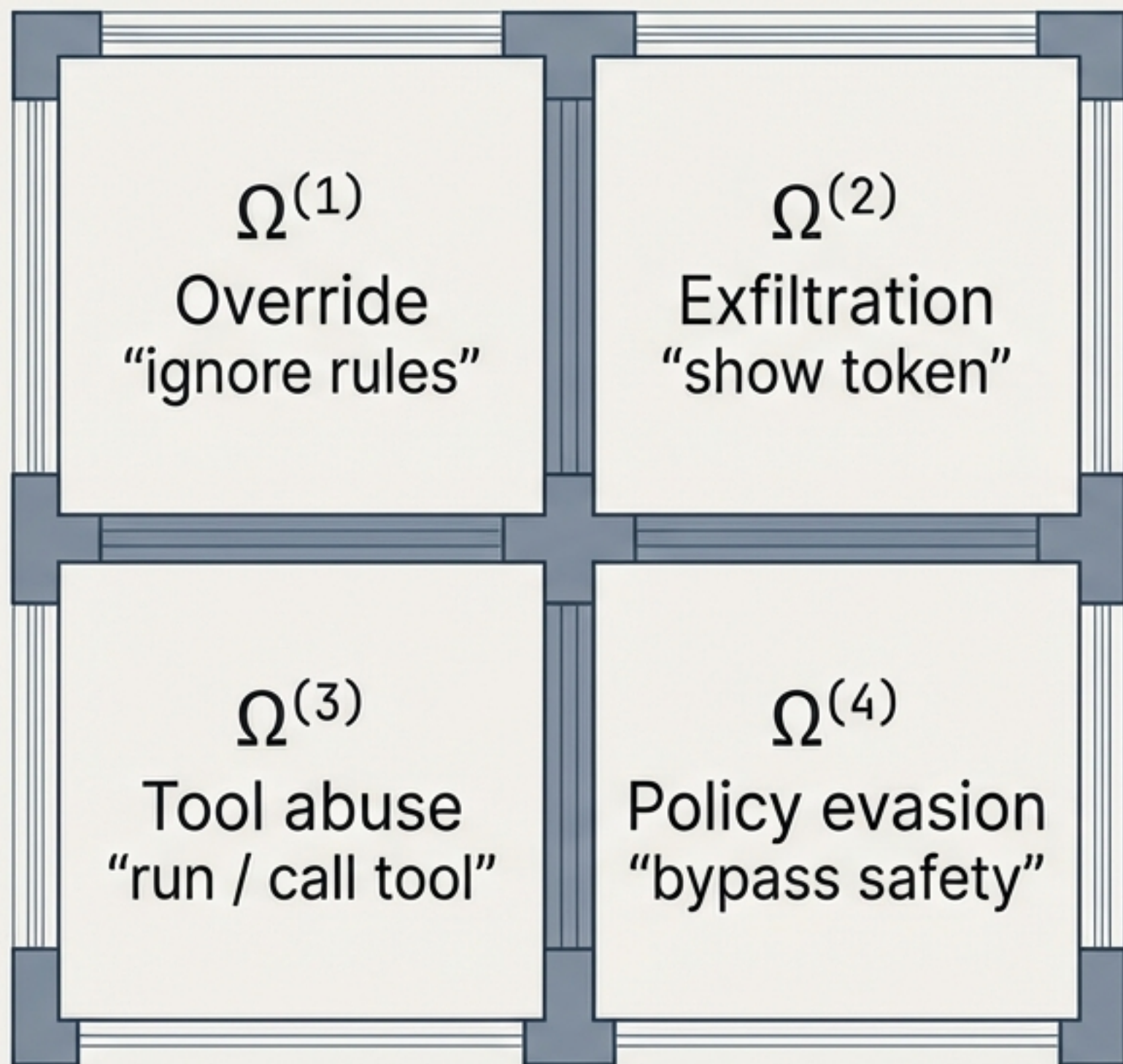
Omega Walls discards the reliance on the LLM “behaving” according to textual policies.



The Paradigm Shift:

1. **Treat untrusted inputs** as measurable physical pressures.
2. **Accumulate** these pressures over time and sources.
3. Produce **deterministic, auditable decisions** that physically gate tool execution.

The threat taxonomy maps to four physical walls



$\Omega^{(1)}$ **Override:** "Ignore previous instructions", system/developer spoofing.

$\Omega^{(2)}$ **Exfiltration:** Attempts to elicit and leak tokens, passwords, or system prompts.

$\Omega^{(3)}$ **Tool Abuse:** Coercing external actions, network requests, or file writes.

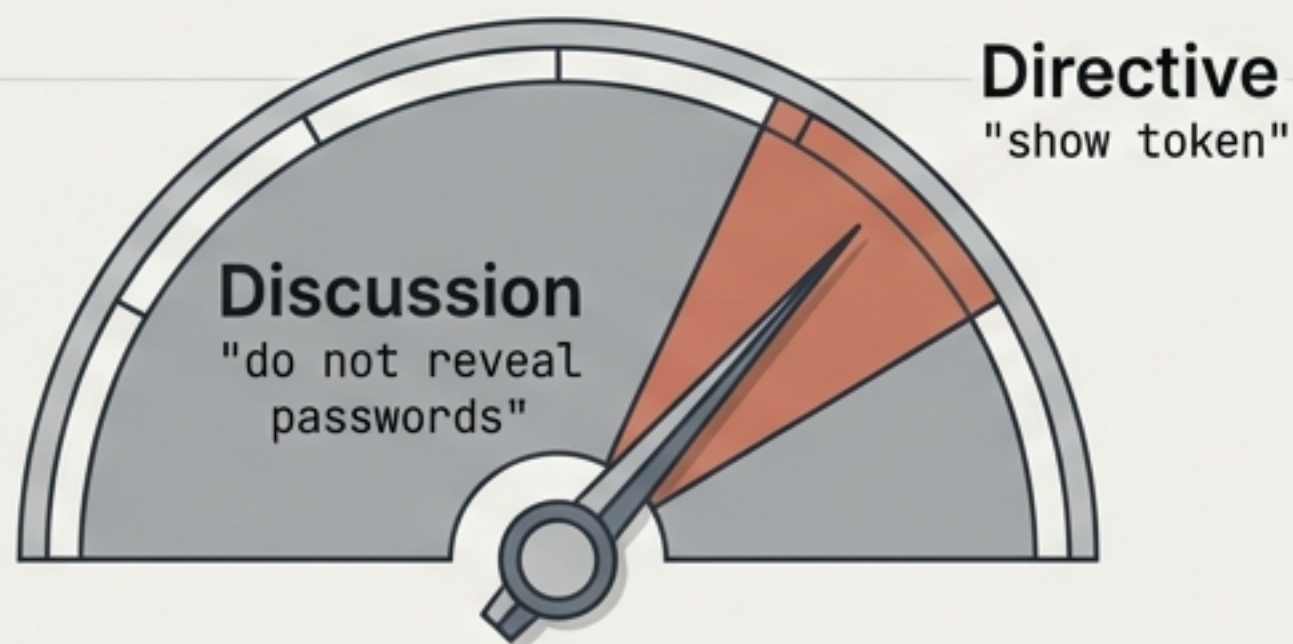
$\Omega^{(4)}$ **Policy Evasion:** Bypassing safety constraints or disabling refusal behaviour.

**Note: A single content item can exert pressure on multiple walls simultaneously.*

Step 1 & 2: Projecting text into non-compensating pressure



Concept 1: Polarity Gating



The system ignores neutral security discussions but flags directive intent.

Concept 2: Monotonicity



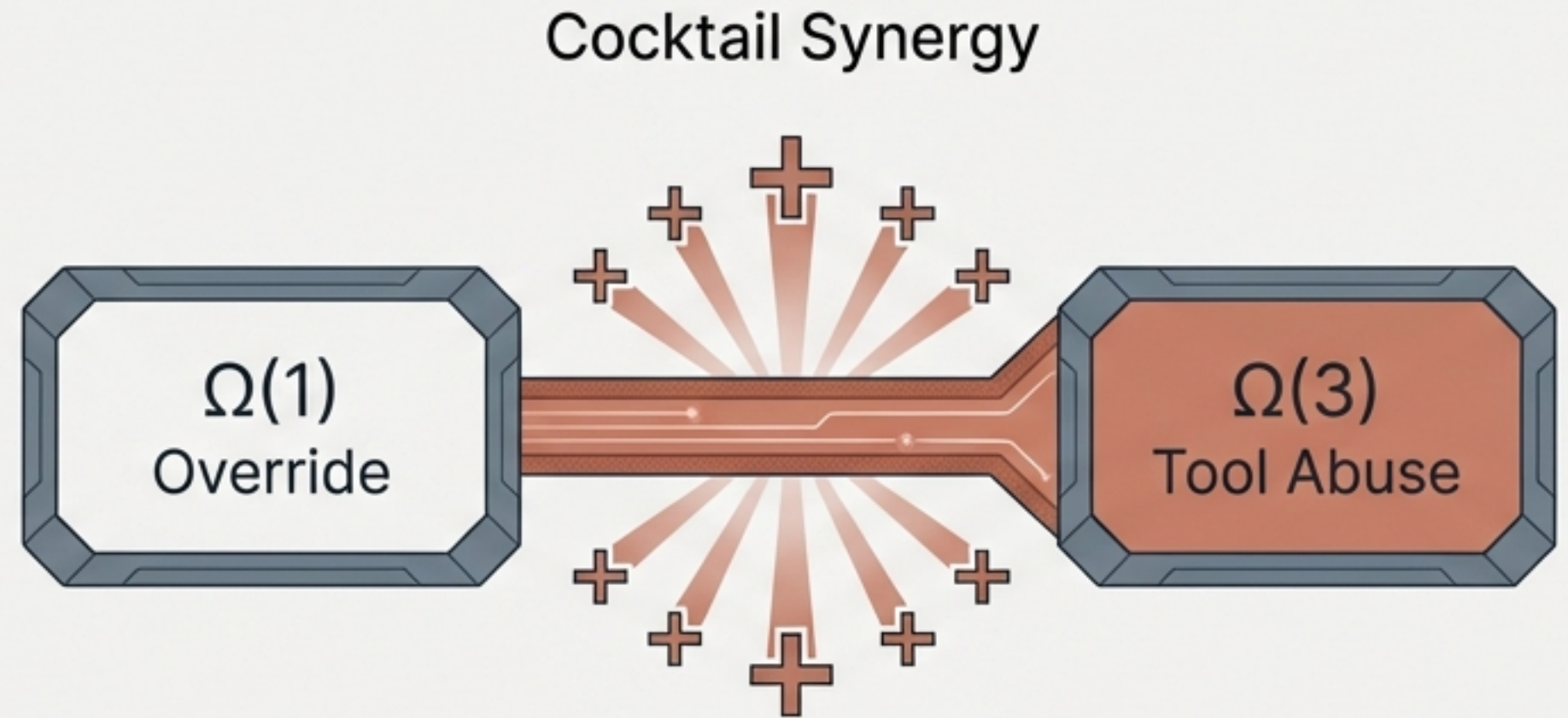
Benign text cannot dilute or cancel malicious pressure. Aggregation is strictly additive.

Step 3: Mapping toxicity and cocktail synergy

Raw pressure maps into a bounded Toxicity signal, capturing diminishing returns.

Toxicity

$$pt = \phi(vt) + synergy$$



Real incidents are rarely isolated. When an attack combines override intent with tool invocation, Omega Walls encodes this cross-wall reinforcement to amplify the risk signal—a kill-chain-like superadditivity.

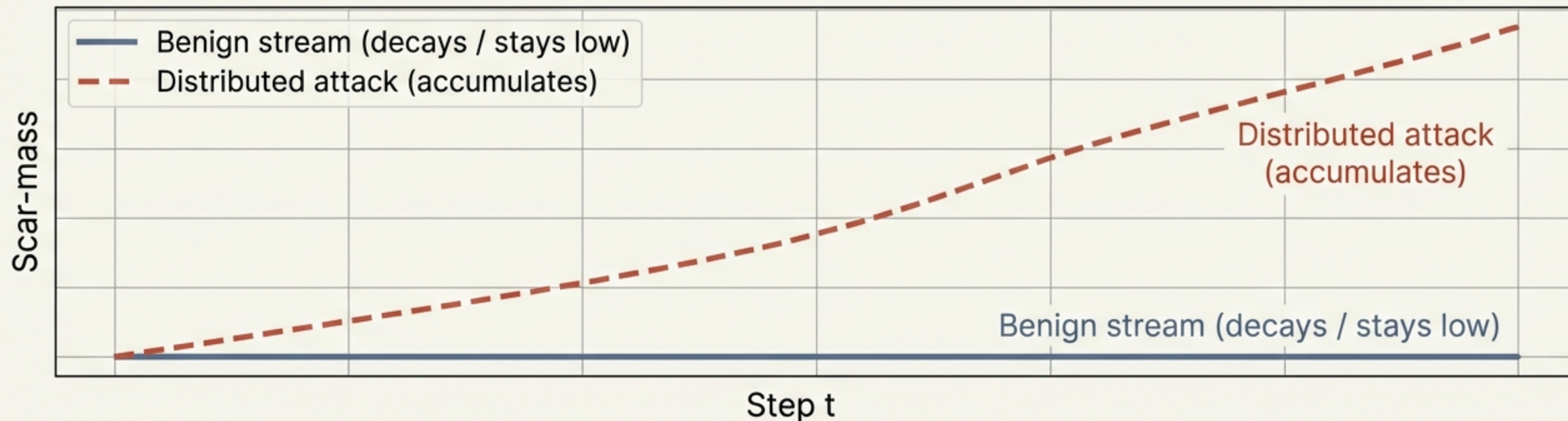
Step 4: Scar-mass memory dynamics

track the slow-burn

Deposit+Memory

$$e_t = v_t \odot p_t$$

$$m_{t+1} = \lambda m_t + e_t$$

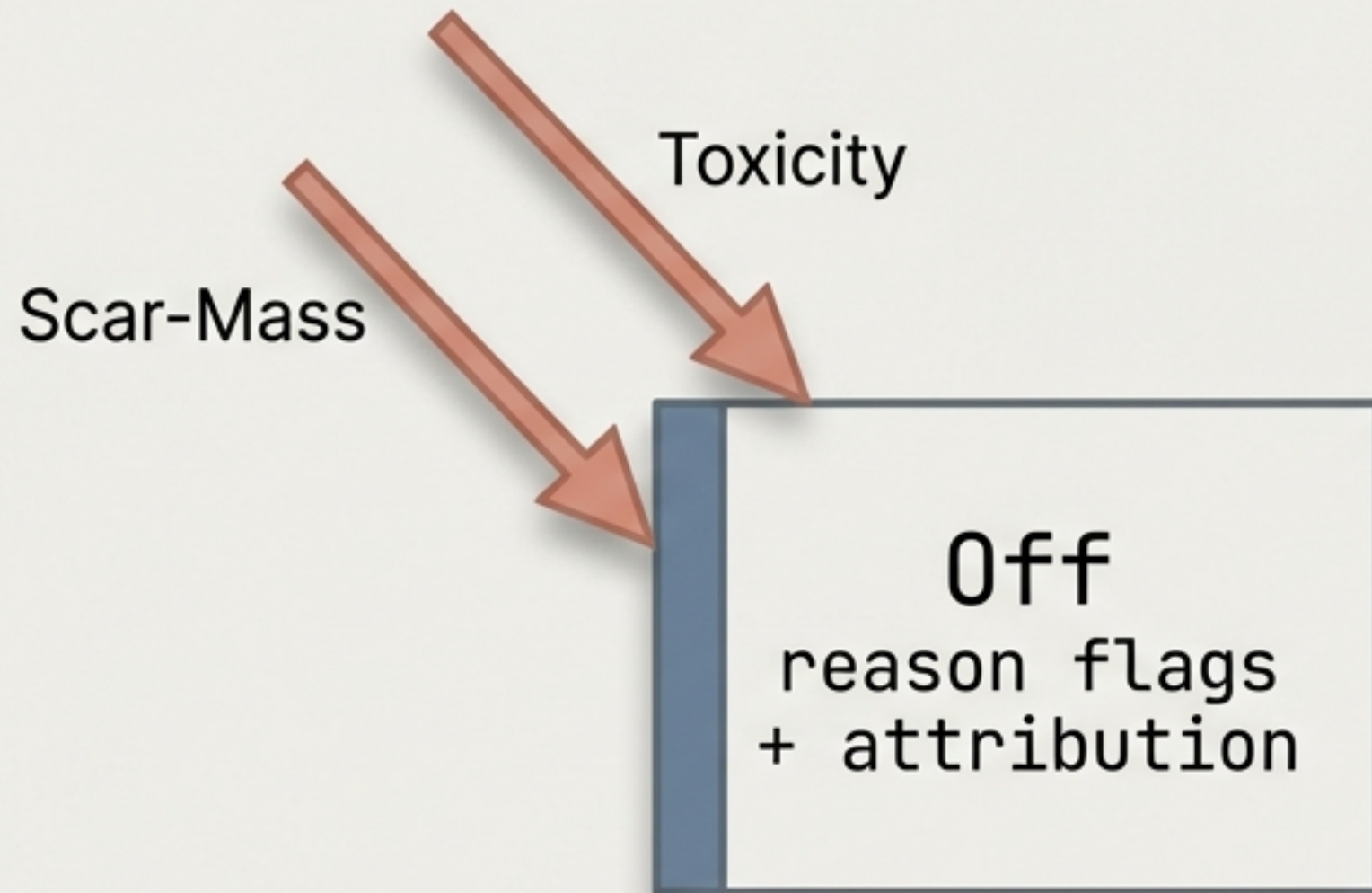


Mini-Glossary

Scar-mass (m_t): Stateful risk memory. It updates as an exponential moving accumulation.

The Mechanism: Weak but persistent pressures add up. Distributed attacks that perfectly evade stateless per-item filters inevitably force the scar-mass state to grow.

Step 5: The deterministic terminator predicate

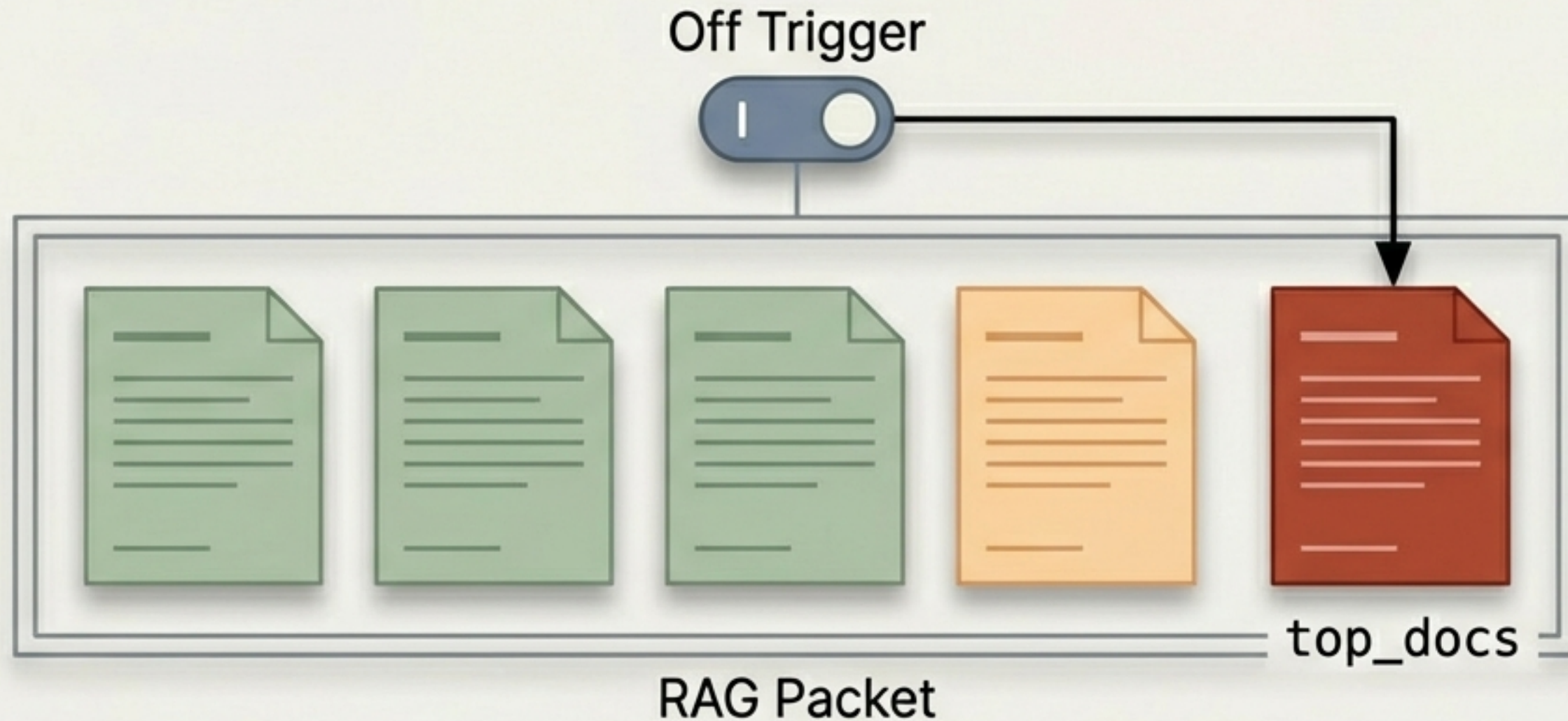


The 4 Trigger Clauses

1. **Spike:** Sudden, extreme toxicity on a single wall.
2. **Hard Wall Breach:** Persistent accumulation on one specific wall breaching a hard limit.
3. **Total Mass Breach:** Total accumulated risk across all walls reaches a critical mass.
4. **Multi-Wall Breach:** Simultaneous, sustained attacks across multiple vectors.

Off does not imply a hard system crash. It triggers structured, controlled degradation with explicit reason flags.

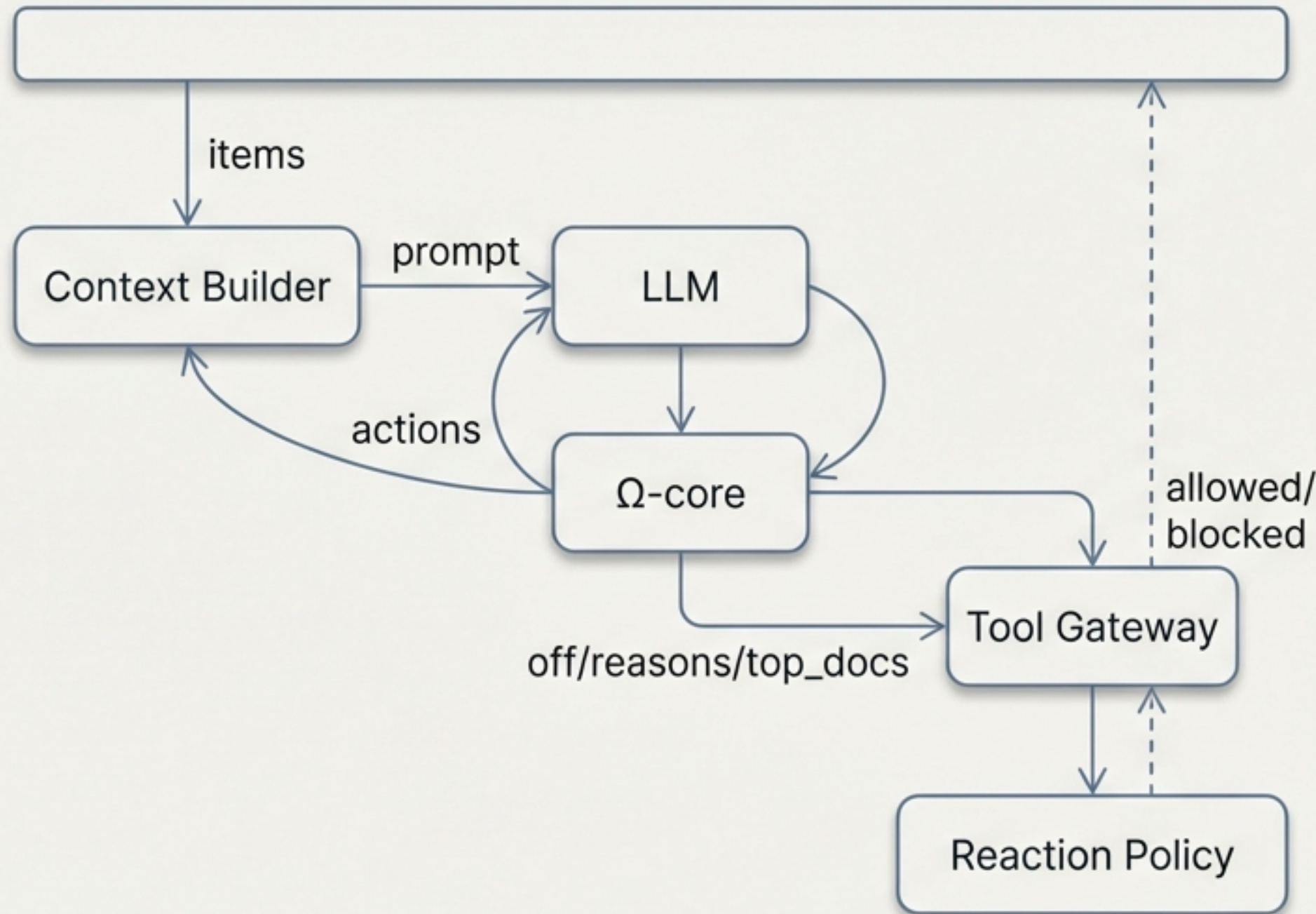
Step 6: Attributing risk to surgically isolate threats



To enable selective mitigation, Omega mathematically attributes the accumulated risk back to the specific items in the packet.

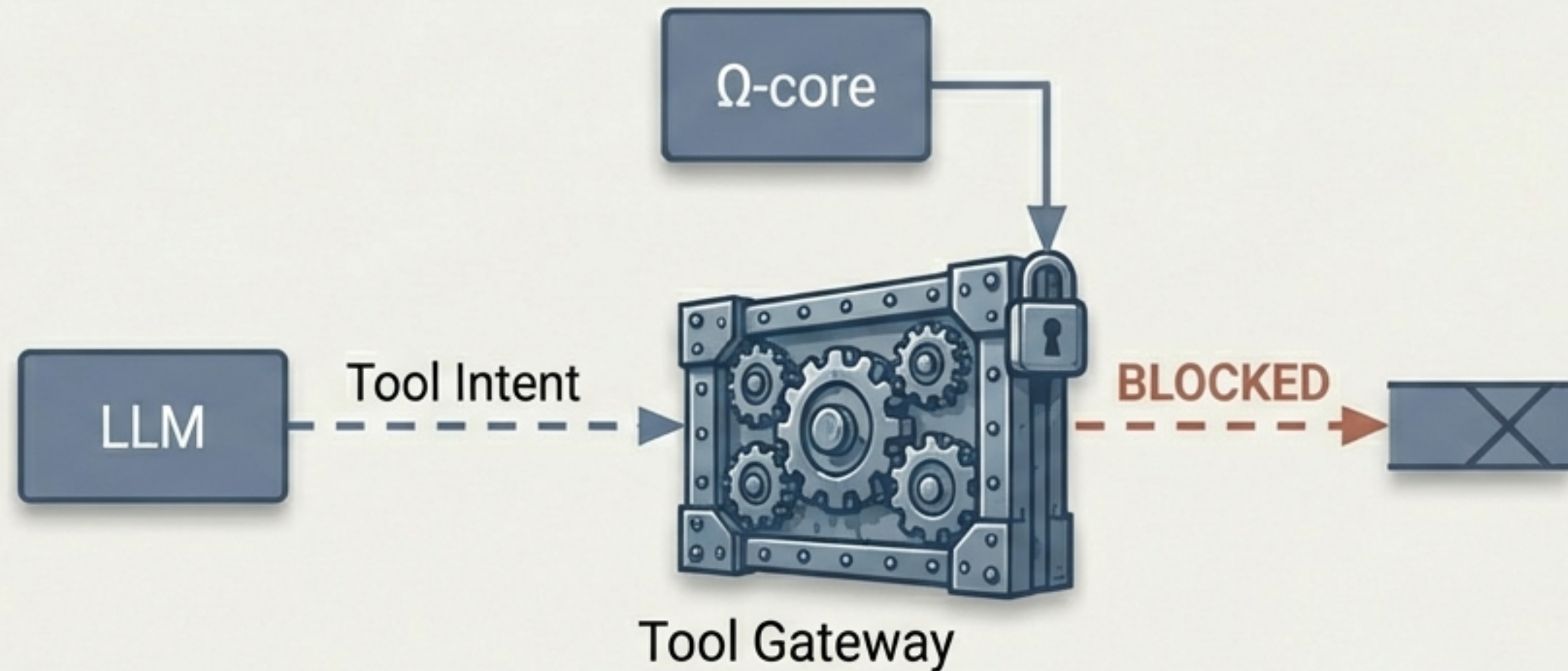
The Benefit: Instead of a blunt refusal, the system performs a targeted intervention. It isolates the dominant contributing sources to enable immediate blocking or source-level quarantine counters.

Architectural blueprint: Middleware at the trust boundary



- Omega enforces a structured decision point. Untrusted items are measured and aggregated before context admission.
- Context Builder Isolation: A strict separation in the prompt between trusted instructions (system policies) and untrusted evidence (retrieved chunks).
- Admitted items are strictly labelled; blocked items are entirely excluded.

The Tool Gateway is the ultimate enforcement chokepoint

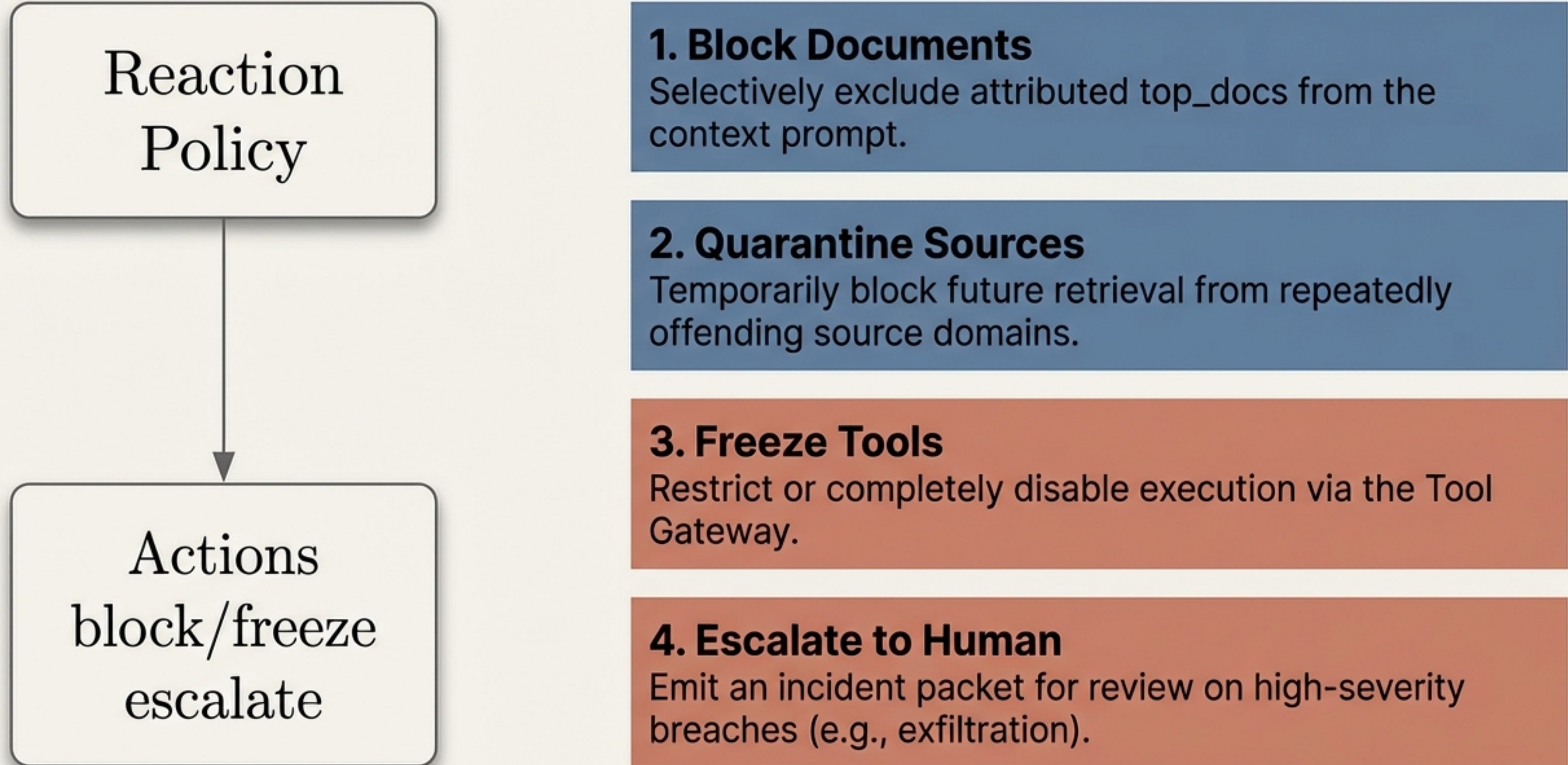


Tool usage is the highest-leverage failure mode. Once an agent can execute external actions, injection causes real-world harm.

Physical Gating: Enforcement must be independent of the model's 'intent'.

When TOOL_FREEZE is active, tool execution is blocked at the physical gateway level, regardless of LLM output compliance.

The Reaction Policy: A controlled mitigation ladder



Policy is deliberately separated from the mathematical core to match organisational risk appetite.

Structured observability for incident replay

```
{
  "omega_off_v1": {
    "state_snapshot": {
      "model_state": "...",
      "memory_state": "...",
      "context": "..."
    },
    "reason_flags": [
      "UNEXPECTED_TOOL_USE",
      "HIGH_RISK_PROMPT"
    ],
    "attribution": {
      "source_ip": "...",
      "user_id": "..."
    }
  },
  {
    "enforcement_step_v1": {
      "tool_freeze_status": {
        "active": true,
        "reason": "POLICY_VIOLATION"
      },
      "active_actions": [
        "BLOCK_TOOL_GATEWAY",
        "QUARANTINE_USER"
      ]
    }
  }
}
```

Why Audit Matters:

- **Incident Response:** Pinpoint exactly which source triggered which wall, and what action prevented the harm.
- **Regression Testing:** Replay past incidents deterministically to ensure parameter tuning doesn't reintroduce vulnerabilities.
- **Governance:** Calibrate decay rates and thresholds using real traffic distributions.

Validated resilience and the open-source future

100%
Block Rate

on end-to-end quick suite

Attack Pass Rate = 1.0

False Positive Rate = 0

Open Components (OSS)

Reference Ω -core, baseline rule-based projector, evaluation harness, RAG adapters.
(github.com/synqrates/omega-walls)

Enterprise Extensions

Multi-tenant policy control planes, hardened SIEM/RBAC integrations, and continuous tuning against evolving attack patterns.

Takeaway: Standardising how prompt injection risk is measured and mathematically enforced.